STAFF REVIEW · H446 298 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. When a base case is met.

2. A partial choice/state considered in the search.

3. It avoids exploring states proven unable to succeed.

4. 12>10 ⇒ go right to15; 12<15 ⇒ insert as left child of15.

I Do — source worked steps

1. Predict first — The first path over-shoots: 2+3+4=9, so reject that branch.

2. Explain the method — Undo 4, skip 3, then include 4 → 6.

We Do — guided CHECK

1. Predict first
Including 1 and 3 produces 4; adding 4 overshoots 5.
Explain the method
Undo 4, skip 3, then include 4 → 5.

You Do — source mark schemes

1. Try subset [2,3,4] [1], sum9 [1], prune as >6 for non-negative inputs [1], undo last choice [1], exclude 3 [1], choose 4 with 2 for sum6 [1].

2. Check total==target return true [2]; end-of-input false [1]; prune total>target [1]; include branch [2]; exclude branch [1]; combine with OR [1].

3. Rows/columns/diagonals constrained [2]; choose next row/column then recurse [1]; undo when no legal square [1]; worst-case combinatorial branching [1]; safe placement checks no shared column or diagonal [1].

Exit Ticket — source answer

1. True; reversing a choice lets the search explore another branch.